Republic of the Philippines
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COLLEGE *of* COMPUTER STUDIES

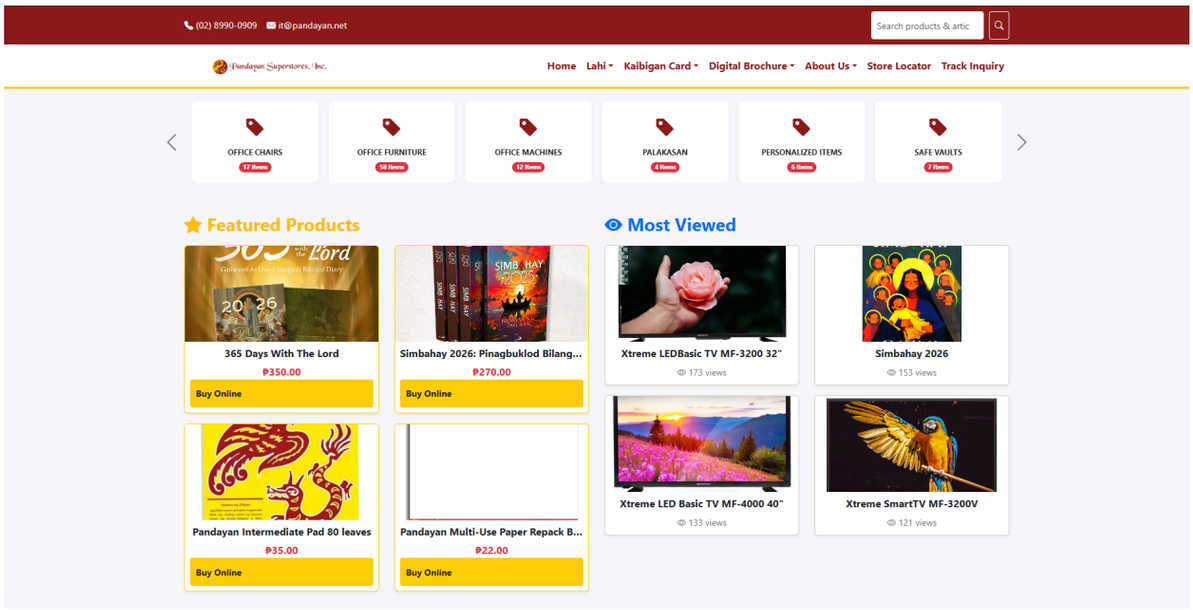
CS 3215 - Human Computer Interaction

LEARNING TASK (DESIGN THINKING PROCESS)

GROUP NO: **9** SECTION: **BSCS3A**

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SYSTEM/APPLICATION: *Pandayan Bookshop E-Commerce Website (Pandayan Bookshop)*



SCENARIO OR PROBLEM: ***Pandayan Bookshop,*** a trusted Filipino educational supplies retailer with a strong physical presence, currently operates an e-commerce website that falls short of meeting modern user expectations and digital shopping standards. Through preliminary analysis and observation, here are primary users recurring problems identified:

- The current website, as you enter the link in some e-commerce website one thing for it to be called, shopping online is with an icon of a cart, but their website lacks a seamless shopping experience with no shopping cart visibility, unclear and chop product images and information.
- Students and teachers struggle to quickly find specific school supplies and educational materials due to missing products like essential school and education supplies which should be the focus of the website which leads to cart abandonment as it does not focus much on these essentials.
- Despite its growing popularity of website e-commerce in the Philippines along with Lazada and Shopee, Pandayan website feels poor, and abandon as its product when buying online is just linked and redirected to Lazada, Shopee or By inquiring at the nearest main Pandayan bookshop per Main Branch of Region.
- The current website also has some unusable tab features that when clicked are getting a website error (example is when clicking view all categories of their digital brochure).

- The website updates depends on events as it follows realtime events or recent or upcoming (example this February is Valentines Month) but the featured products don't follow the trend.
- Pandayan is known for its discount and points giving but in their e-commerce website the Loyalty program disconnect: Kaibigan Card members couldn't see their points balance or available discounts while shopping.
- The website checking availability when inquiring about products lacks a nearest Store Locator of the nearest Pandayan Shop in your Area instead it will always show the nearest Main Branch.

With this, the website lost revenue opportunities from users who prefer online shopping but find the current platform inadequate and feel abandoned as their site has many unclear engagement to users. The need for redesign is critical to transform Pandayan Bookshop's online presence into a competitive, user-friendly platform that serves today's digital needs of Filipino students, teachers, and institutions, especially their high quality focused retailer of school and office supplies while maintaining the brand's trusted reputation in educational supplies.



EMPATHIZE

Actions:

To gain an empathetic understanding of Pandayan Bookshop's websites and things to enhance, we conducted comprehensive analysis of the website ourselves, navigating through all major sections to experience firsthand the user journey from landing page or homepage to product interfaces to checkout tab and latest news and events interface as well its hypertext reference attribute on their website to see it its working and what it does. We documented our observations, disappointment, and pain points as we attempted to complete common tasks such as searching for specific school supplies, browsing product categories, events and news, and understanding why there is no cart icon and checkout process directly onto their website.

We and the team members shadowed ourselves as users interacting with the Pandayan website, observing how long it took to find specific products, noting where confusion occurred, and identifying points where we felt disconnected and feel that it is seamlessly lacking. We also gather and attempt to try the website with the same tasks (example clicking on one embedded hyperlink reference) to see if it has the same response to all users and also gain insights into how different types of users(students and teachers) experience the platform.

With Philippines one of the growing e-commerce users, we decided also to conduct a competitive analysis by interacting with other similar e-commerce websites, including major platforms like National Bookstore Online, LCC Mall Shop Online as well as leading platforms like Shopee and Lazada. We compared their features, navigation from their homepage, search functionality, responsiveness, checkout processes, and other features to find out, so that we can identify what Pandayan's website was missing and what worked well in these competing platforms.

We also analyzed reviews and feedbacks by reading reviews, comments on Pandayan social media pages (Facebook) as well as customers review on their redirect linked in from the web page to shopee or lazada where it was found that they were been frequently rated highly for the authenticity of their product to the picture posted in the website. We also reviewed the website's other existing features other than shopping such as the Kaibigan Card program, Digital

Brochure section, search function, forms, and its navigation locator to understand how these elements were integrated into the online shopping experience.

Key Insights:
Through our hands-on exploration and analysis, we discovered several critical insights about user needs and pain points. When attempting to find specific products, it sometimes gives results that are inexact to what is being searched (the algorithm is confused since some of the user search does not exist so it gives a search that is close-by on those search words.) One thing we notice when viewing from Laptop/Desktop view vs Phone view it appears to be primarily designed for desktop viewing, making it extremely difficult to use on mobile devices where most users prefer to shop.

Another function we notice about the website is that it is now being much more anchored as a “Digital Brochure” rather than a true e-commerce platform. While some products display “Buy Online” buttons, the shopping cart functionality is not clearly visible and is persistently felt missing throughout the browsing experience. This creates confusion about whether users can actually complete purchases online through and within the website or if they need to be redirected to another website or visit a physical store or call to order. The Kaibigan Card loyalty program, which appears to be a significant part of Pandayan’s business, is mentioned on the website but not integrated into online shopping experience. Users that are members on this cannot see their points, access exclusive discounts or feel their membership value doesn’t adds up when shopping online.

Altogether, we experienced disconnection when products appeared to be featured but clicking through provided minimal information or unclear purchasing paths. We felt uncertainty about whether the website was meant for browsing or just an actual online shopping. When using mobile devices, the experience was much disconnected and at the same time poor for some interface that we instinctively wanted to switch to desktop or just visit the nearest physical store instead which defeats the purpose of having an e-commerce website. However, we also felt the need to build a connection within this website by redesigning and doing suggested right improvements, this platform could genuinely serve users.

DEFINE

Actions:
We analyzed the research data to pinpoint the main problem. The goal was to define a clear problem statement that focuses on the user’s needs. Gathering the data from our synthetic interview with user personas helped us see the pain points through their specific perspectives.

- Lawrence (User Persona):
  - Goal: He wants to buy specific art materials for valentine’s day
  - Frustration: He searches the website, finds the item he needed, but upon clicking “Buy” he is redirected to Shopee, where the shipping takes 3 to 5 days. He leaves the website to find a physical store near him instead.

- Lowi (User Persona):

- Goal: He wants to buy 100 notebooks to restock his store and use his Kaibigan Card points to save money.
- Frustration: He cannot log in to check his points on the website. He feels that buying on some online shopping platforms is more expensive because he misses out on the membership discounts and points.

We found out that the main challenge is the “disconnection.” The physical Pandayan store is known for being helpful through community initiatives, and fostering a positive environment, but the website feels empty and broken. The users expect the website to be an extension of the physical store, and not just a list of links to other apps.

Problem Statement:
“The community needs a centralized, fully functional online store that connects directly to their nearest local branch inventory and loyalty account.”
This is because of the current system of redirecting to third-party apps like Lazada and Shopee. It removes the convenience of quick local delivery and prevents loyal customers from using their hard-earned Kaibigan Card points, leading them to abandon their online carts.

IDEATE

Actions:
We held a brainstorming session to generate our proposed solutions. We used the "How Might We" method to turn the problems into opportunities for design. We focused on three main questions:

- How might we make the website feel as friendly and helpful as the physical store?
- How might we let users use their loyalty points online without complications?
- How might we fix the "out of stock" anxiety that teachers feel when ordering in bulk?

Potential Solutions:
- Native Shopping Cart: A built-in cart system where users can review items, adjust quantities, and see the total price immediately.
- Integrated Checkout: A secure payment page that accepts local payment methods (GCash, Maya, Cash on Delivery) directly on the site, removing the need to switch apps.
- Smart Search Bar: A search tool that suggests products as you type (e.g., typing "ballpen" shows specific brands like Pilot or Faber-Castell) to help students find items faster.
- Kaibigan Card Dashboard: A dedicated "My Account" section. When a user logs in, they see their current points balance. During checkout, they can click a simple "Apply Points" button to get a discount.
- "Pick Up in Store" Option: Since Pandayan has many branches, we added an option at checkout to "Click and Collect." This allows students to pay online and pick up the item at the nearest branch on the same day, saving on shipping fees.

PROTOTYPE

Actions:

To demonstrate the cogency of our design, we have developed an interactive online model of the redesigned Pandayan Bookshop E-Commerce Website. The prototype is aimed at addressing the significant concerns that were raised during the Define stage, specifically, the absence of a native shopping experience, the integration of loyalty programs, and inefficient mobile usability.

Figma was our main tool of prototyping since it enables us to iterate quickly, create interactions that are clickable and can share the prototype to be tested by a user. The prototype will be mid-fidelity, as it will have realistic layouts, navigation flow, and core interface elements, but will not have full backend functionality.

The design of the prototype was made mobile-first, followed by desktop views, to consider the present generation use of e-commerce platforms by most Filipino users.

Prototype Description:

The prototype will be a simulation of a complete functional Pandayan online store and not an online brochure. It has the following major screens and features:

1. **Homepage**
    a. Shopping cart icon located on the upper navigation bar.
    b. Good concentration on school and education supplies (notebooks, ballpens, art supplies).
    c. Products appealing to seasonal requirements (e.g. Valentine art supplies, back to school packs, etc).
    d. Smart suggestions on search bar.

2. **Product Detail Pages and Product Listing**
    a. Organized categories.
    b. Good quality and clean product images.
    c. Clear stock availability, clear availability of prices and branch availability.
    d. Instead of redirect links, there is an Add to Cart button.

3. **Native Shopping Cart**
    a. Managers are able to change amounts and automatically view new balances.
    b. High visibility call-to-action check out button.
    c. Perfect shopping cart presence on every page.

4. **Checkout Flow**
    a. Having integrated check out on the site.
    b. There are GCash, Maya, and Cash on Delivery as payment options.
    c. Store Pickup with a user-chosen nearest branch.

5. **Kaibigan Card Dashboard**
    a. Login page for those who availed the Kaibigan Card membership.

b. Highlighted point balance and available discounts.
c. Application of points at checkout.
d. Strengthens the contribution of loyalty membership to the internet.

6. **Store Navigation**
   a. Relies on location selection to reveal the closest Pandayan branch to the user.
   b. Highlighted stock availability on each branch.

The prototype is a direct response to the main problem of disconnection in that it converts the website of Pandayan into a centralized e-commerce site. Customers can now shop, shop, buy without leaving their apps to third party applications. The experience resembles the usefulness and convenience of the brick and mortar store and matches the modern online shopping requirements.

TEST

Actions:
To measure the prototype effectiveness, we have performed usability testing to 6 participants that will be the key user base of Pandayan:

- 4 students
- 2 educators / small businesses customers

Every participant was requested to fill in the following tasks using the prototype:

- Find a specific school supply.
- Add items to the cart.
- Check Kaibigan Card points.
- Checkout items via delivery or store pickup.

Our usability testing method was task-based, after which the interviews and the feedback questionnaire took place. The participants would be allowed to think aloud as they went through the prototype in order to locate the points of confusion and usability problems.

Feedback and Insights:
What worked well:

- The visible shopping cart was welcomed and it was appreciated immediately by the users.
- The Kaibigan Card Dashboard ensured that the users received a sense of reward and were more willing to make purchases.
- Pick Up in Store minimized the fear of delayed delivery.
- Navigation using the mobile was much easier than in the case of the original site.

What needs improvement:

- There were users who desired more vivid signs of bulk discounting.
- A mapview would be useful in the store locator.
- The process of checkout can be further made to be quicker.

Overall Result:
Everyone managed to accomplish their tasks without any outside assistance. The redesigned platform enabled them to experience more confidence, less confusion, and a desire to do online shopping as opposed to the original site.

Next Steps:

Based on the testing feedback, we have identified several priority improvements to implement before moving forward with development. In the immediate short term, we will refine its mobile capacity by increasing its responsiveness especially when it comes to its image size and adding subtle animation or prompts that draws attention when users first land on the homepage. Another thing to be addressed is the discoverability issue identified with the missing icon cart when it should be online shopping this ensures online shopping functionality doesn't go unnoticed.

We will also enhance the bulk ordering feature's visibility by adding clearly labeled user Orders if it's for schools or offices. This will also include a brief explanation of how buyers can request quotes, access volume discounts, and set up recurring orders for classroom supplies. In addition to this, we will add many items especially the one that belongs in school and supplies as it should be the focus of the website. We plan also to instead of just all buyers redirecting to lazada and shopee we will add also in the form when we want to buy in the nearest pandayan, a nearest locator of the local pandayan store or if can we will just also add a button for directly checking out the product the user purchase through the website this will set user expectations to making the website really a purchase yours truly in the website of pandayan and reduce abandonment of the website.

We are also planning to conduct a round of usability testing with these refinements and refinements that will be added in the future including us and other classmates for more diverse participants to ensure design works for a broader audience giving us fresh perspectives on whether our improvements effectively addressed the initial feedback.

These adjustments and metrics to be measured will be monitored every after effect of changes thereafter, allowing us to continuously refine and optimize the platform based on us acting as real users checking its behaviors and also feedback on what we think needs to be redesigned or to be changed.

TAKEAWAYS

Name of Member: Aquino, Sean Xander B.
Contribution to the Group: Define, Ideate, Leadership and Team Morale
Learnings: We applied our teamwork and cooperation in this exercise of maneuvering through the pain points of the users of the site. This collaboration led to the discovery of many problems and issues daily users have to overcome, including the inability to locate the required materials, being redirected to other online shopping services, and the inability to know how many points they have in their Kaibigan Card, among others. During brainstorming, we realized that our ideas could be taken seriously by the owner himself and this encouraged us to be more critical and creative. This experience helped us understand that innovation cannot necessarily need complicated solutions sometimes, it only needs greater insight into the users and a solution to their most fundamental frustrations. Generally, this exercise has equipped not only our problem-solving abilities but also our communication and support of other colleagues in their thoughts. It reminded us that a user centred design is necessary when designing systems that are not only functional, but also meaningful and convenient to the individuals who use them.

Name of Member: Colico, Carl Renz M.
Contribution to the Group: Scenario, Problem, Empathize, Next Steps, and Team Morale
Learnings: What I experience and learned with this learning task is how there are website launched and appeared to be and should be an online shopping yet the user (us) feel like it is aren't, making me feel uncertain on how whether this website like this Pandayan is really meant for online shopping browsing or just a e-brochure of their items, where I instinctively see it is really lacking when it comes to its function as an e-commerce website it also lacks functionality of being called a online website of pandayan as their is no directly category of where you can buy it through just their website. This suggests that with the right improvements, this platform could genuinely serve Filipino students and teachers better than foreign competitors.

Name of Member: Montenegro, John Renzzo C.
Contribution to the Group: System/Application Suggestion, Prototype, Test
Learnings: This activity has made me learn that redesigning a website is not only to make it look better but to actually address real user issues driven by empathy and systematic design thinking. After conducting the analysis of the present platform of Pandayan Bookshop, we found that the primary problem was the lack of connection between its strong physical presence in the stores and its weak online experience. By clarifying user frustrations, brainstorming about the solutions that could be used, creating a working prototype and carrying out the usability tests, we understood the importance of having a user-centered, seamless, and centralized e-commerce system. The process revealed that useful digital solutions should focus on convenience, clarity, loyalty integration and accessibility, particularly in a rising e-commerce setting such as the Philippines.